

5.2 MOSFET Device Characteristics

5.2.1 Qualitative Theory of Operation

5.2.2 Derivation of IV Characteristic

- Long-Channel MOSFET: Square-Law Theory

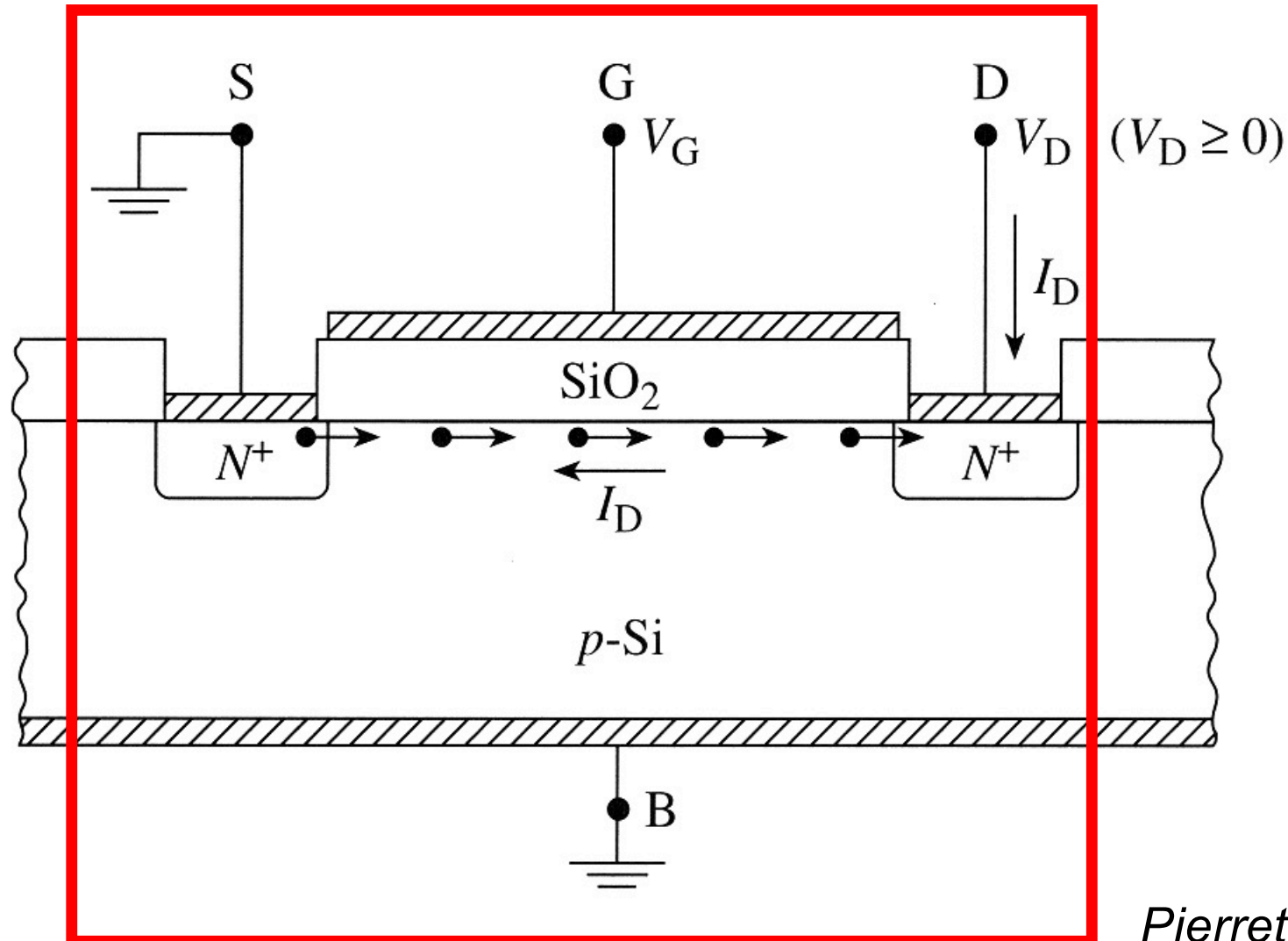
5.2.3 Threshold Voltage

- Workfunction Difference
- Oxide Charges

5.2.4 MOSFET Types and Circuit Symbols

Literature: Pierret, Chapter 17.1-17.2, page 611-630
(for V_T : Pierret, Chapter 18, page 645-681)
Jaeger, Blalock, Chapter 4.2-4.4, page 148-165

N-Channel MOSFET – Device Structure



Pierret, Fig. 17.1

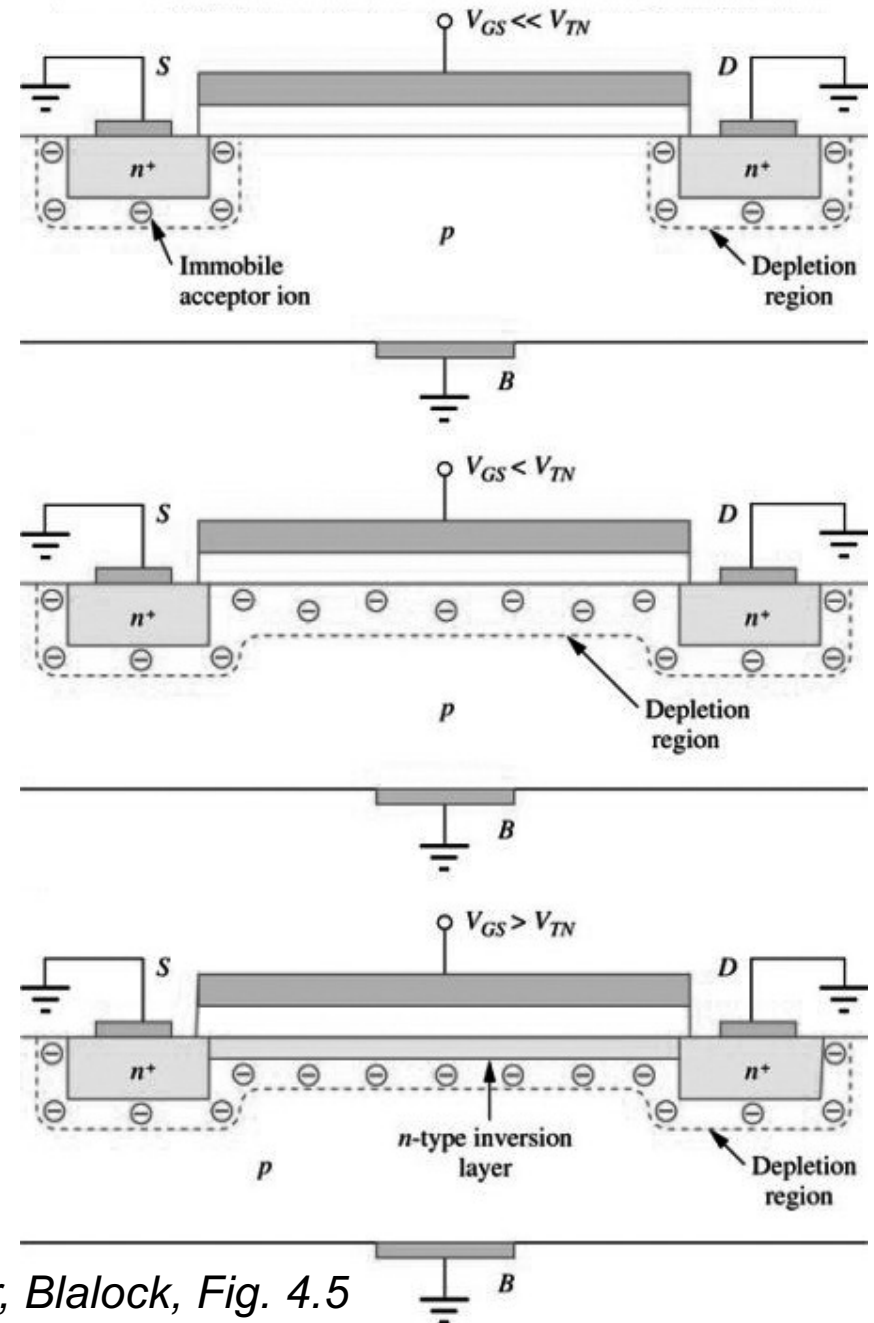
N-Channel MOSFET Device Operation

$V_g < V_T$: Depletion (or Accumulation)

- No conductive path between source S and drain D
- Drain current $I_D = 0$ for all $V_D > 0$

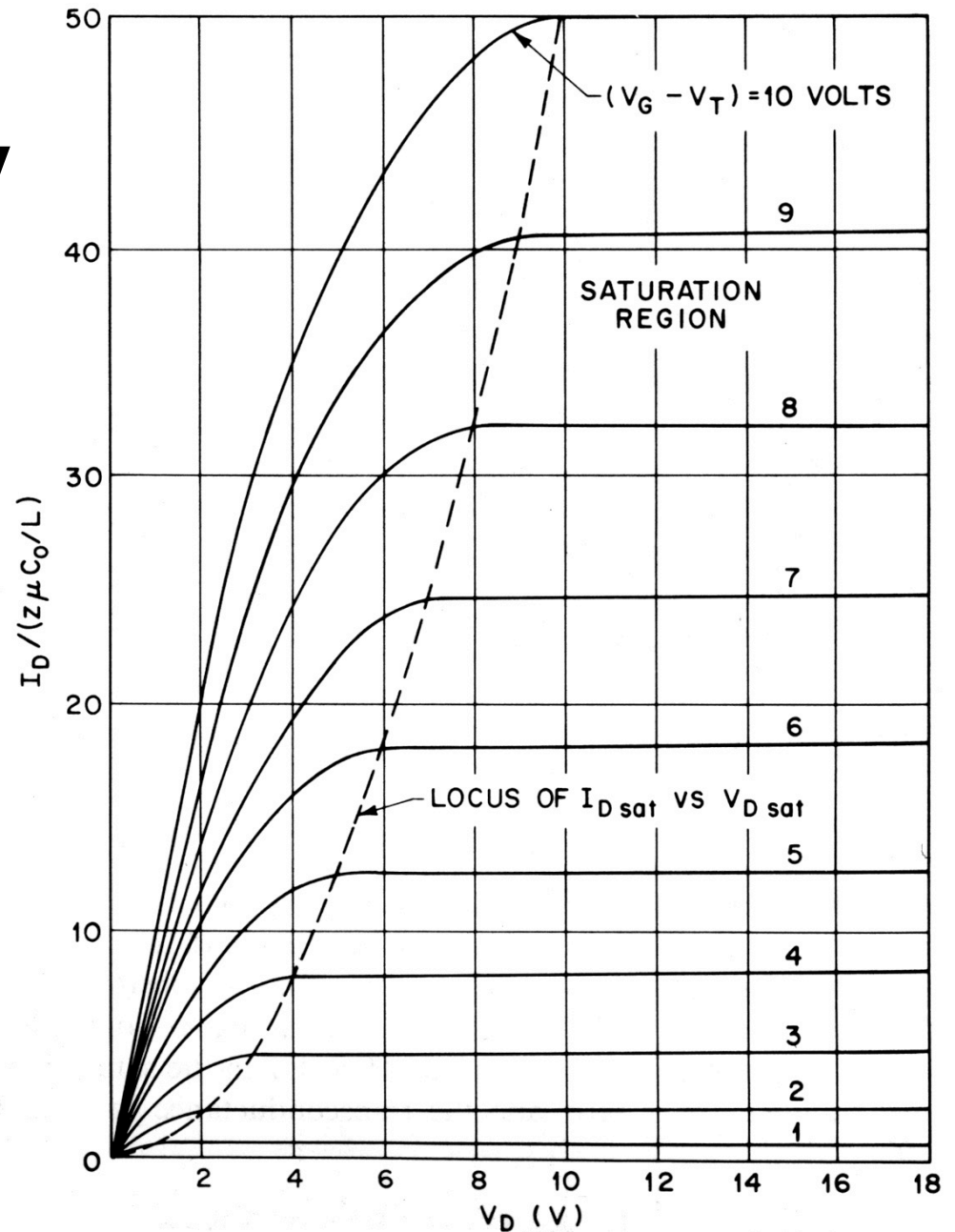
$V_g \geq V_T$: Inversion

- Conductive channel (n-type) between source S and drain D
- Drain current $I_D > 0$ for all $V_D > 0$
- Open Question:
 $I_D(V_D)$ Characteristic?



Jaeger, Blalock, Fig. 4.5

5.2.1 MOSFET Qualitative Theory $I_D(V_D)$ Characteristic



Size, Fig. 6.17

MOSFET – Linear Regime

Linear Regime: $V_g \geq V_T$ and $V_D \ll V_g$

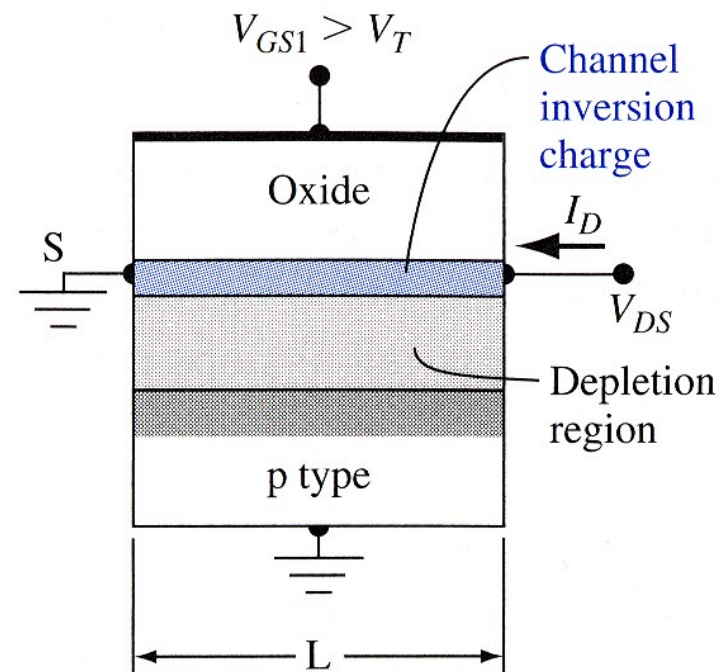
- Width of and carrier concentration in the n-type inversion layer only depend on V_g (and not on V_D), i.e. are constant along the length L of the channel

➔ As a result, the channel resistance R_{channel} does not depend on V_D , but only on V_g

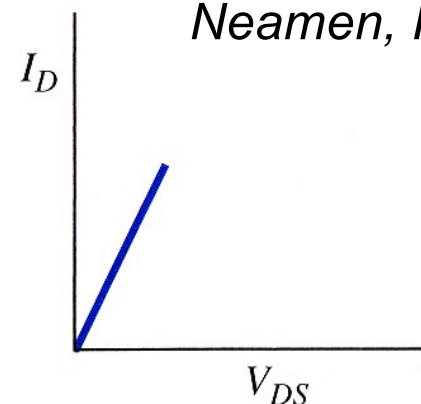
$$I_D = R_{\text{channel}}^{-1} V_D$$

➔ For a given V_g , I_D increases linearly with V_D

➔ R_{channel} decreases with increasing V_g , thus the slope of I_D vs. V_D increases with increasing V_g



Neamen, Fig. 11.40



MOSFET – Non-Linear Regime

Non-Linear Regime

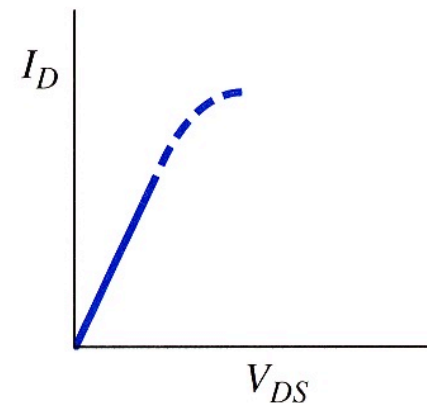
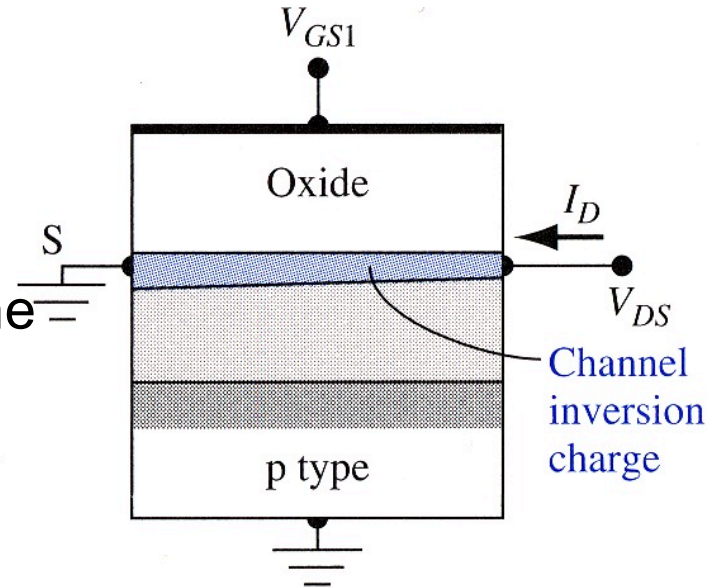
$$V_g \geq V_T \text{ and } V_D < V_{D,\text{sat}}$$

- Width of and carrier concentration in the n-type inversion layer depend on V_g **and** V_D , i.e. are **not** constant along the length L of the channel

➔ As a result, the channel resistance R_{channel} increases with increasing V_D ; effective voltage drop between gate and channel is reduced towards the drain

- Leveling-off of $I_D(V_D)$ characteristic

➔ Note: Jaeger/Blalock call the linear/nonlinear regime the **triode region**



Neamen, Fig. 11.40

MOSFET – Pinch-Off Point

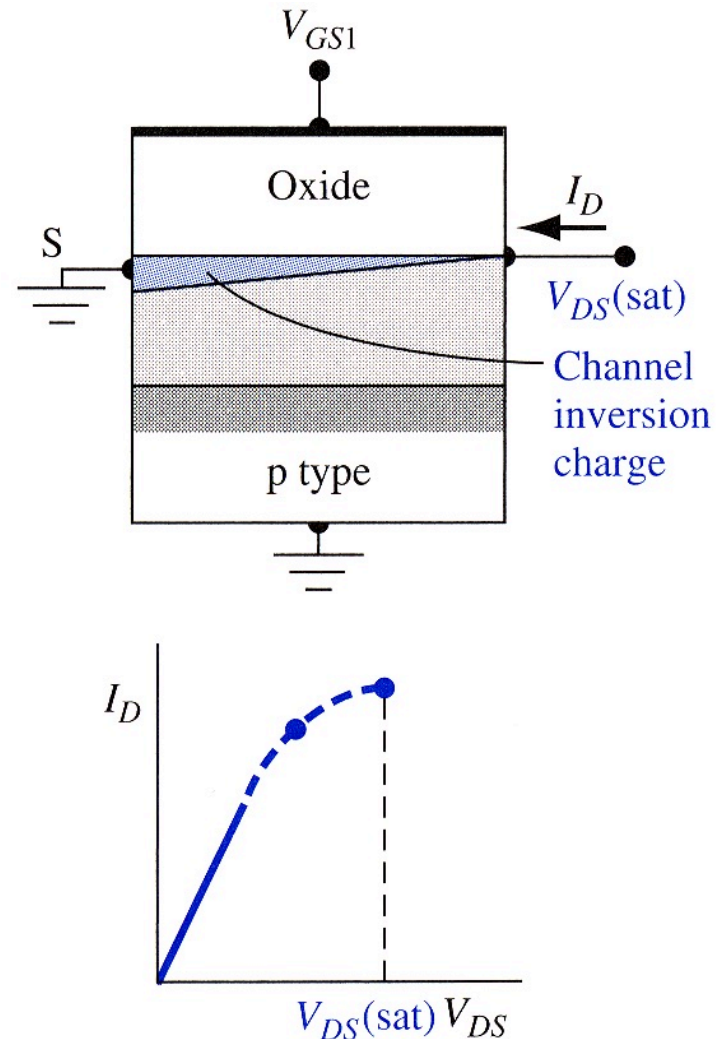
Pinch-Off Point

$$V_g \geq V_T \text{ and } V_D = V_{D,\text{sat}}$$

- The voltage applied across the MOS capacitor at the edge of the drain $\Delta V = V_g - V_{D,\text{sat}}$ is just enough to reach the onset of inversion

$$\left. \frac{dI_D}{dV_D} \right|_{V_D=V_{D,\text{sat}}} = 0 \quad \text{with} \quad V_{D,\text{sat}} = V_g - V_T$$

- This is known as the **pinch-off point**
- V_T is the threshold voltage (the voltage to achieve the onset of inversion) for $V_D = 0$



Neamen, Fig. 11.40

MOSFET – Saturation Regime

Saturation Regime

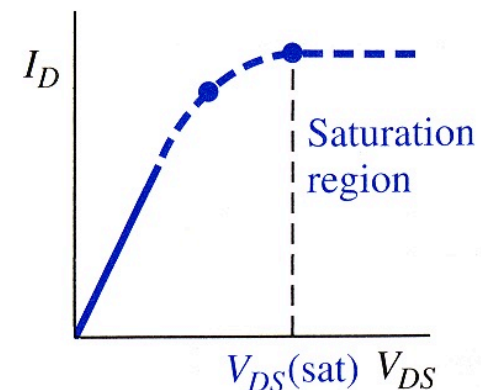
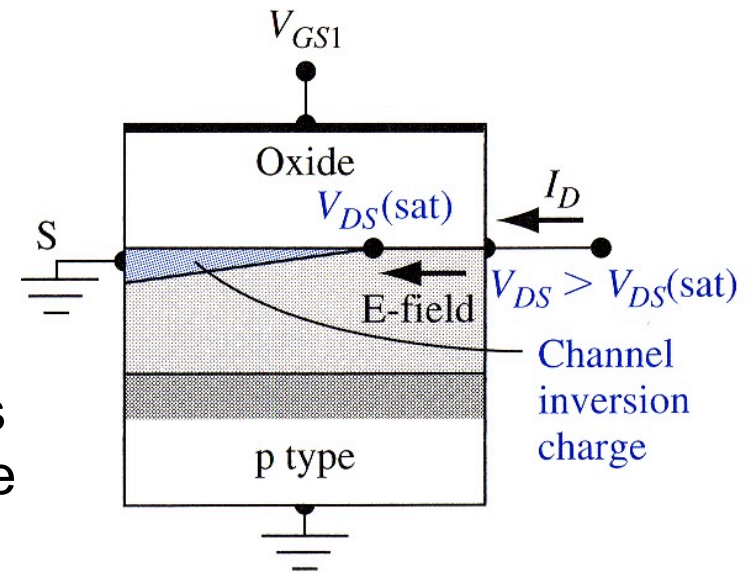
$$V_g \geq V_T \text{ and } V_D \geq V_{D,\text{sat}}$$

- Pinched-off portion of the channel widens (ΔL in Pierret, Fig. 17.2); the pinched-off region is depleted, i.e. it has few carriers and thus a low conductance

➔ As a result, most of the drain voltage in excess of $V_{D,\text{sat}}$ is dropped across a small ΔL :

➔ $I_D = \text{const.}$ for $V_D \geq V_{D,\text{sat}}$ if $\Delta L \ll L$ (long channel approximation)

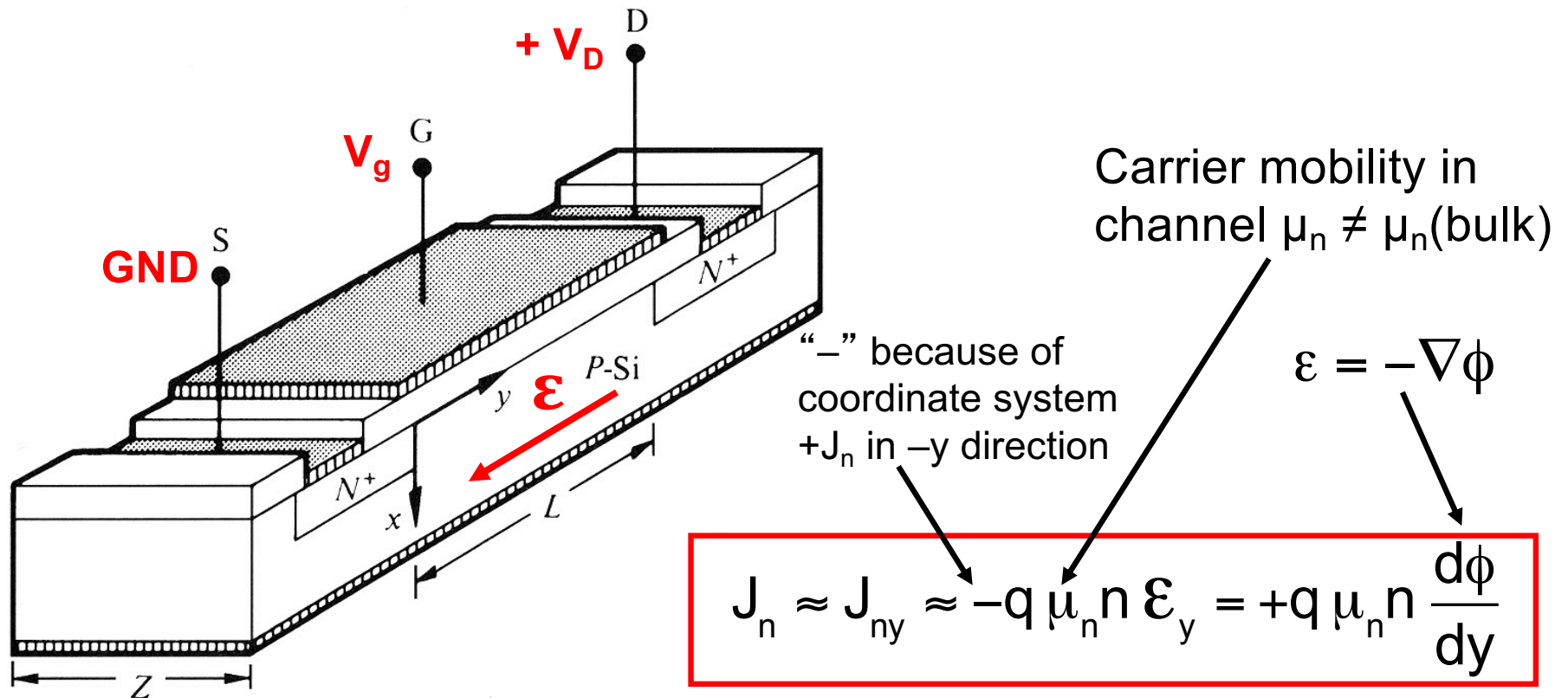
- At the pinch-off point, the carriers are injected into the depletion region and are swept by the strong electric field to the drain contact



Neamen, Fig. 11.40

5.2.2 MOSFET – IV Characteristic

- Assumption:
 - N-channel MOSFET
 - Drift current dominating in channel



Pierret, Fig. 17.6

Note: μ_n and n depend on x and y coordinate

MOSFET – IV Characteristic (cont.)

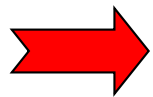
- Current flow is restricted to surface channel

$$I_D = \iint J_{ny} \, dx \, dz = Z \int_0^{x_c(y)} J_{ny} \, dx = -Z \frac{d\phi}{dy} (-q) \underbrace{\int_0^{x_c(y)} \mu_n(x,y) n(x,y) \, dx}_{\equiv \bar{\mu}_n(y) Q_n(y)}$$

Assumption:
 ϕ does not depend on x

Effective
mobility

Charge in inversion
layer at y [C/cm²]



$$I_D = -Z \bar{\mu}_n(y) Q_n(y) \frac{d\phi}{dy}$$

- Integrate over the channel length:

$$\int_0^L I_D \, dy = -Z \int_0^{V_D} \bar{\mu}_n(y) Q_n(y) \, d\phi = -Z \bar{\mu}_n \int_0^{V_D} Q_n \, d\phi$$

Assumption:

$$\bar{\mu}_n(y) = \bar{\mu}_n$$

MOSFET – IV Characteristic (cont.)

$$\Rightarrow I_D = - \frac{Z \bar{\mu}_n}{L} \underbrace{\int_0^{V_D} Q_n d\phi}_{=?}$$

- **Assumption:** Charges added to the metal electrode after the onset of inversion ($V_g \geq V_T$) result in added mirror charges in the inversion layer only, leaving the depletion layer unchanged (see delta-depletion theory of MOS-C): **Square-Law Theory**

$$\Delta Q_{\text{gate}} = -\Delta Q_{\text{semi}} = -Q_n$$

$$\underbrace{\Delta Q_{\text{gate}}}_{[\text{C/cm}^2]} \approx \bar{C}_o \Delta V_g = \underbrace{\bar{C}_o}_{\equiv C_o / A} (V_g - V_T) = \frac{K_o \epsilon_0}{x_o} (V_g - V_T)$$

$$\Rightarrow Q_n = - \frac{K_o \epsilon_0}{x_o} (V_g - V_T)$$

MOSFET – IV Characteristic (cont.)

- Effective potential drop across the MOS-C along the channel length L is $[V_g - \phi(y)]$

$$Q_n(y) = -\frac{K_o \epsilon_0}{x_o} (V_g - V_T - \phi(y))$$

- Inserting in the drain current equation yields

$$I_D = -\frac{Z \bar{\mu}_n}{L} \int_0^{V_D} Q_n d\phi = -\frac{Z \bar{\mu}_n}{L} \int_0^{V_D} -\frac{K_o \epsilon_0}{x_o} (V_g - V_T - \phi) d\phi$$

$$I_D = \frac{Z \bar{\mu}_n \bar{C}_o}{L} \left[(V_g - V_T) V_D - \frac{V_D^2}{2} \right]$$

for $0 \leq V_D \leq V_{D,sat}$ and $V_g \geq V_T$

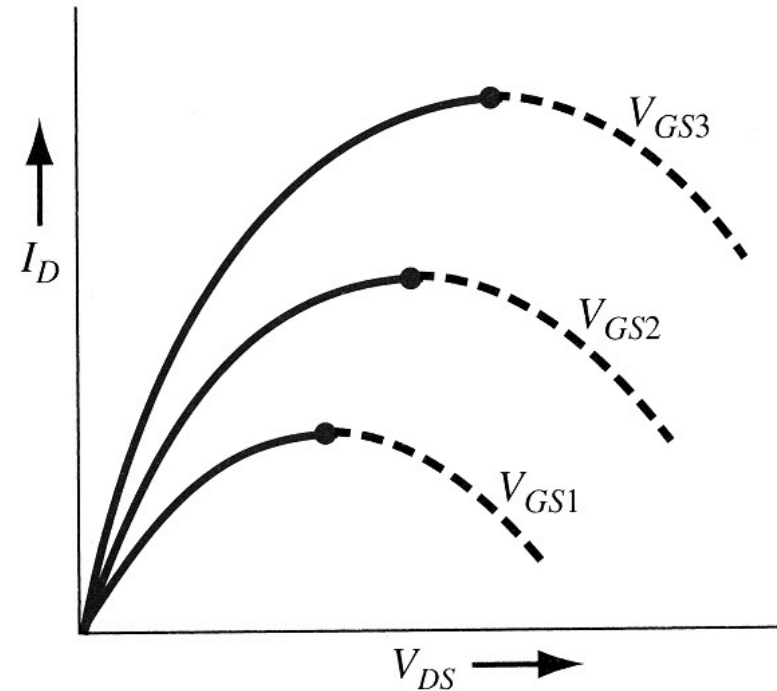
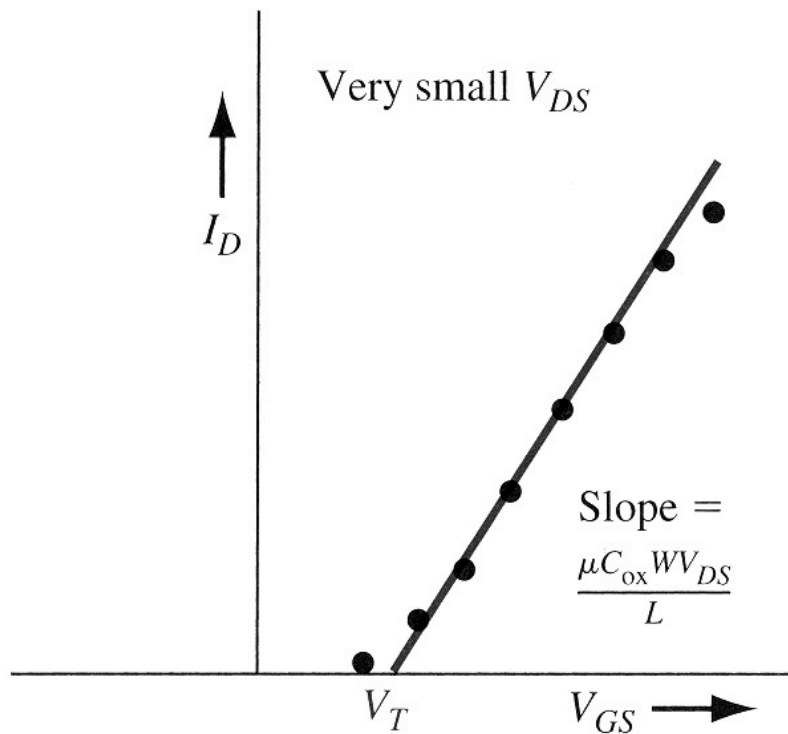
Square-Law Theory

$$V_{D,sat} = V_g - V_T$$

$$I_{D,sat} = \frac{Z \bar{\mu}_n \bar{C}_o}{2L} (V_g - V_T)^2$$

MOSFET – IV Characteristic

– Square-Law Theory –



Neamen, Fig. 11.48
and Fig. 11.49

$$I_D = \frac{Z \bar{\mu}_n \bar{C}_o}{L} \left[(V_g - V_T) V_D - \frac{V_D^2}{2} \right]$$

MOSFET – IV Characteristic

– Square-Law Theory –

- **Linear/Non-Linear Regime (Triode Region)**

$$0 \leq V_D \leq V_{D,\text{sat}} = V_g - V_T$$

$$I_D = \frac{Z \bar{\mu}_n \bar{C}_o}{L} \left[(V_g - V_T) V_D - \frac{V_D^2}{2} \right]$$
$$I_D \approx \frac{Z \bar{\mu}_n \bar{C}_o}{L} \left[(V_g - V_T) V_D \right] \quad \text{for } V_D \ll V_g - V_T$$

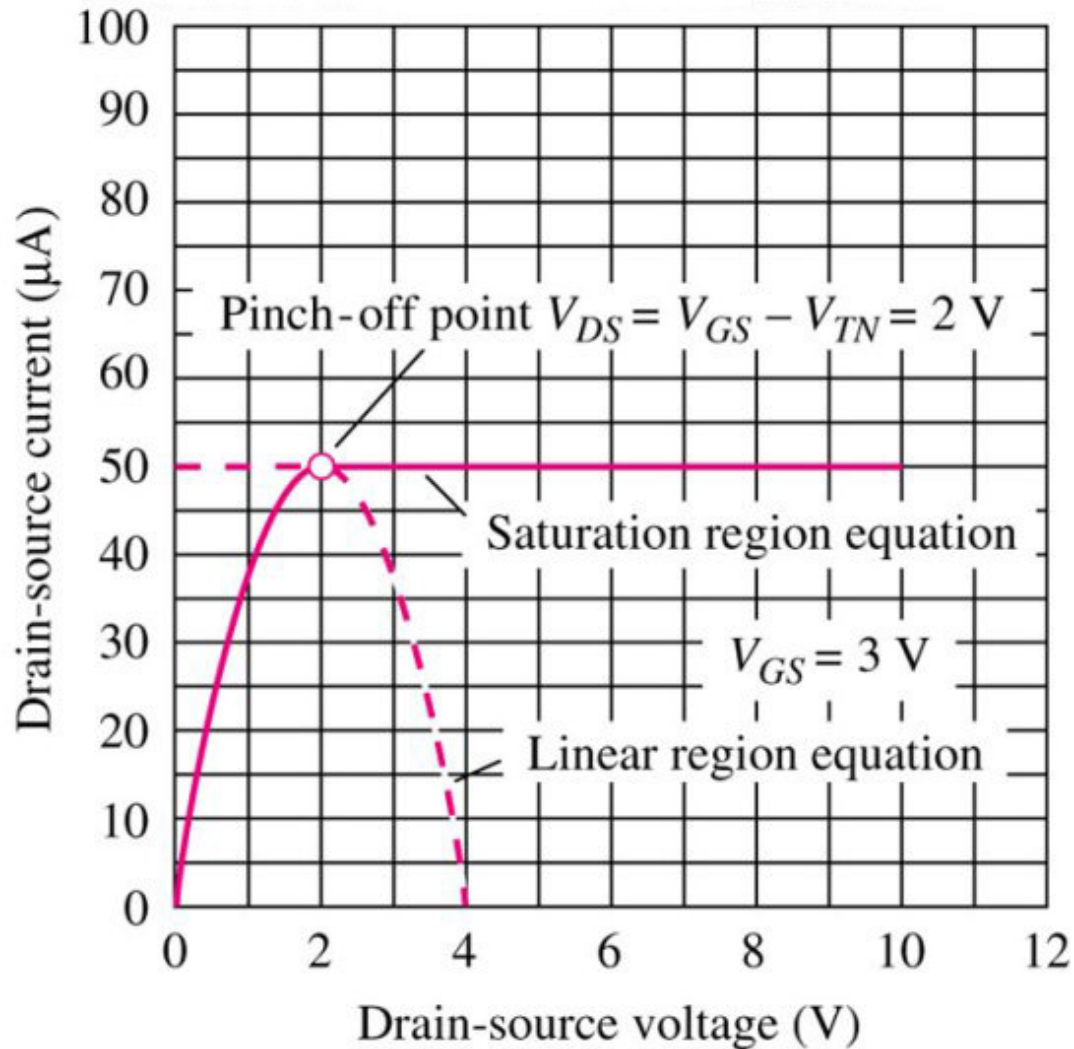
- **Saturation Regime (Pinch-Off Region)**

$$V_D > V_{D,\text{sat}} = V_g - V_T$$

$$I_D = \frac{Z \bar{\mu}_n \bar{C}_o}{2L} \left[(V_g - V_T)^2 \right]$$

MOSFET – IV Characteristic

– Square-Law Theory –



Jaeger, Blalock, Fig. 4.10

MOSFET – IV Characteristic

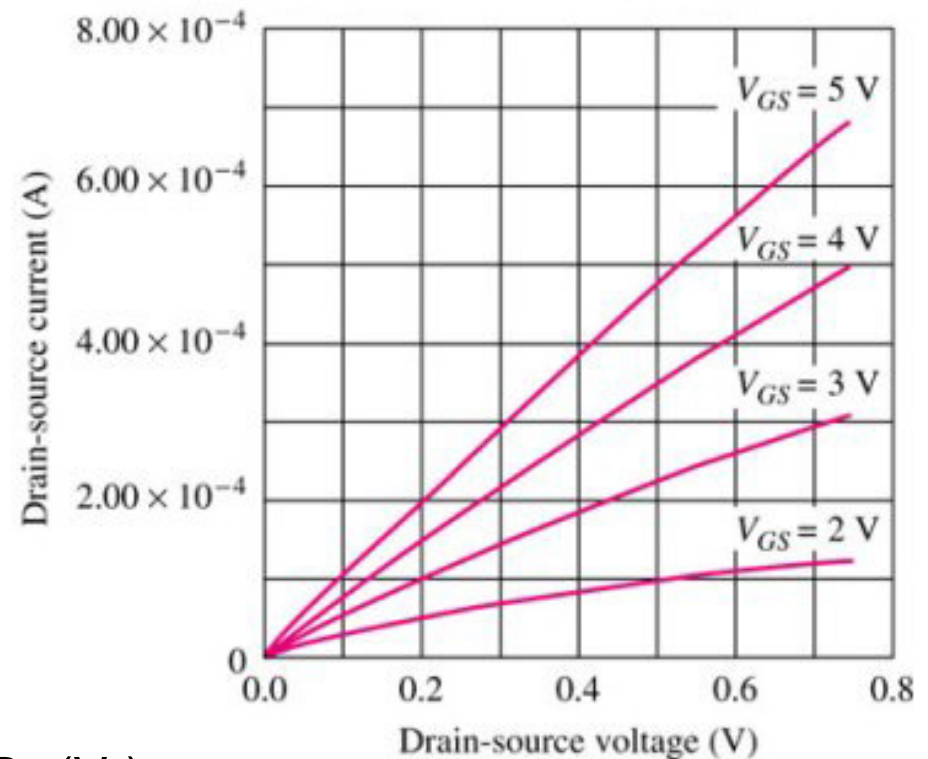
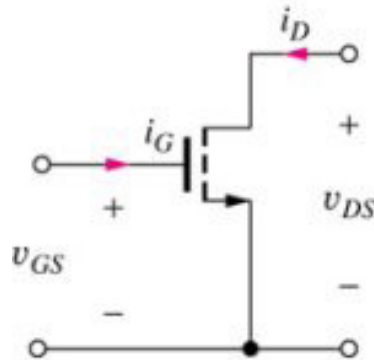
– Linear Regime –

- NMOS device (n-channel MOSFET) in common-source configuration operated in linear regime

- Example: $V_T = V_{TN} = 1 \text{ V}$

$$K_n = \bar{\mu} \bar{C}_O = 250 \mu\text{A} / \text{V}^2$$

Jaeger, Blalock,
Fig. 4.7



$$I_D \approx \frac{Z \bar{\mu}_n \bar{C}_O}{L} [(V_g - V_T) V_D]$$

$$R_{on} = \left[\frac{\partial I_D}{\partial V_D} \bigg|_{V_D \rightarrow 0} \right]^{-1} = \left[\frac{Z \bar{\mu}_n \bar{C}_O}{L} (V_g - V_T) \right]^{-1} = R_{on}(V_g) \quad \leftarrow \text{Voltage-Controlled Resistor}$$

MOSFET – IV Characteristic

– Limitations of Square-Law Theory –

1. Depletion layer width depends on V_g and V_D , i.e. $W = W(y)$; the square-law theory assumes that all added charges are added to the inversion layer; in reality, some of these **charges are added to the depletion layer**; thus, I_D is always overestimated in the square-law theory: use **Bulk-Charge Theory** instead!
2. The drain current is assumed to be zero for $V_g < V_T$; in reality, a channel exists even for $V_g < V_T$, resulting in a small **sub-threshold current**
3. The **drain current does not self-saturate** in the square-law (and bulk-charge) theory; the saturation in the pinch-off portion ($V_D > V_{D,sat}$) of the IV characteristic is “artificially” constructed

Use **Charge-Sheet** or **Exact-Charge** theories

5.2.3 Threshold Voltage

– No Work Function Difference –

- Assuming $\Phi_M = \Phi_S$ and an **n-channel MOSFET**, the threshold voltage is identical to the MOS-C gate voltage at the onset of inversion ($\phi_s = 2 \phi_F$)

$$V_T = 2\phi_F + \frac{K_s x_o}{K_o} \sqrt{\frac{4qN_A}{K_s \epsilon_0} \phi_F} = 2\phi_F + \overbrace{\frac{x_o}{\epsilon_0 K_o}}^{=C_o^{-1}} qN_A \sqrt{\overbrace{\frac{4\epsilon_0 K_s}{qN_A} \phi_F}^{=W_T}}$$

$$V_T = 2\phi_F + \frac{qN_A W_T}{C_o} \equiv 2\phi_F - \frac{Q_B}{C_o}$$

with the charge per area $Q_B = -q N_A W_T$ in the depletion region at the onset of inversion

- Similarly, the threshold voltage for a **p-channel MOSFET** is ($Q_B = +q N_D W_T$)

$$V_T = 2\phi_F - \frac{K_s x_o}{K_o} \sqrt{\frac{4qN_D}{K_s \epsilon_0} (-\phi_F)} = 2\phi_F - \frac{Q_B}{C_o}$$

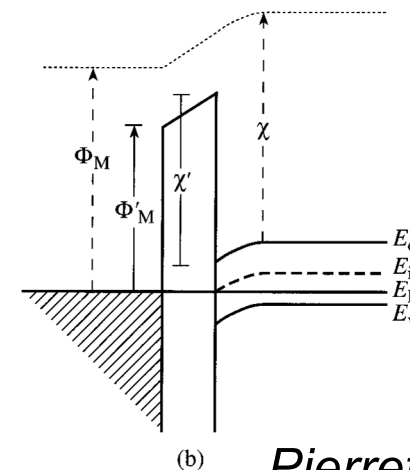
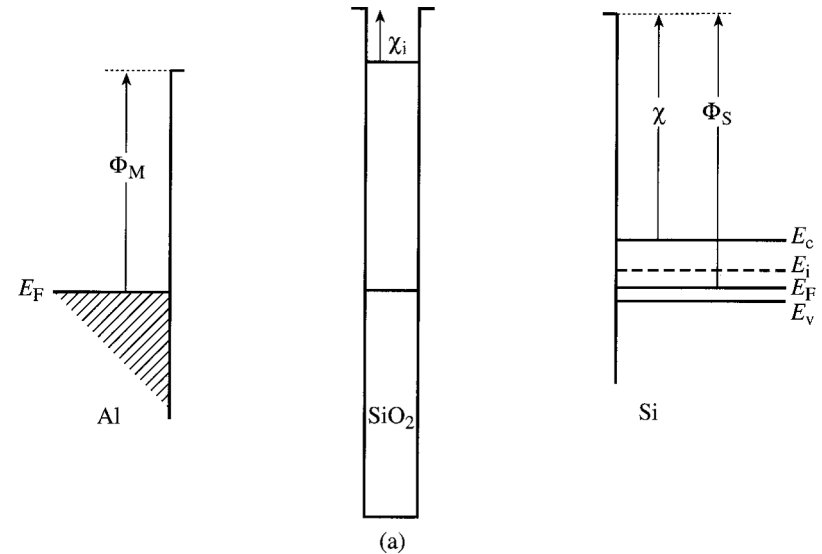
Threshold Voltage

– Work Function Difference ϕ_{MS} –

- In reality $\Phi_M \neq \Phi_S$, i.e. a voltage $\phi_{MS} = q^{-1} (\Phi_M - \Phi_S)$ has to be applied to the gate to achieve flat band condition
- As a result, the threshold voltage is shifted by the work function difference:

$$V_T = \phi_{MS} + 2\phi_F - \frac{Q_B}{C_o}$$

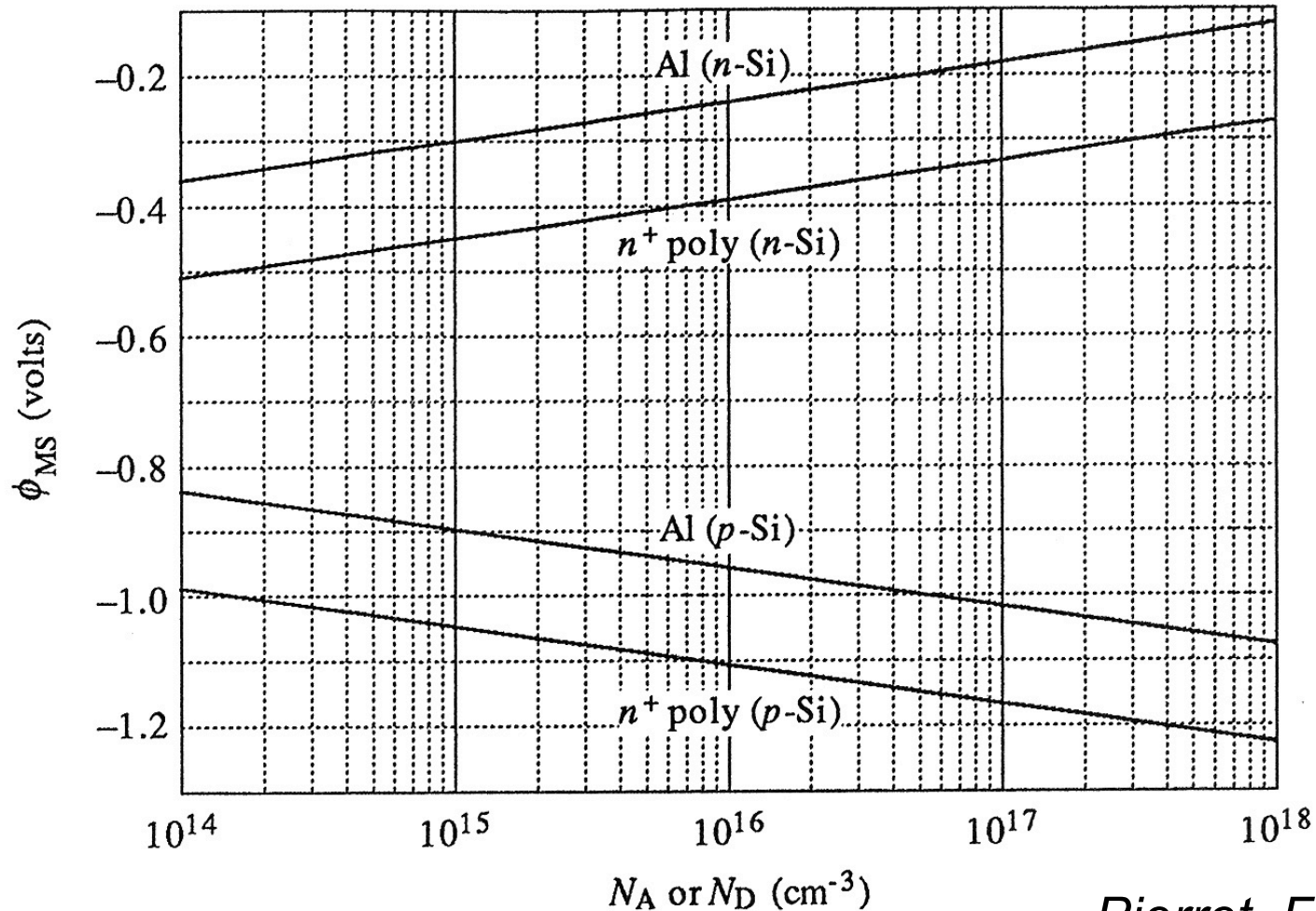
	n-channel	p-channel
ϕ_F	> 0	< 0
ϕ_{MS}	< 0	< 0
Q_B	< 0	> 0



Pierret, Fig. 18.1

Threshold Voltage

– Work Function Difference ϕ_{MS} –



Pierret, Fig. 18.3

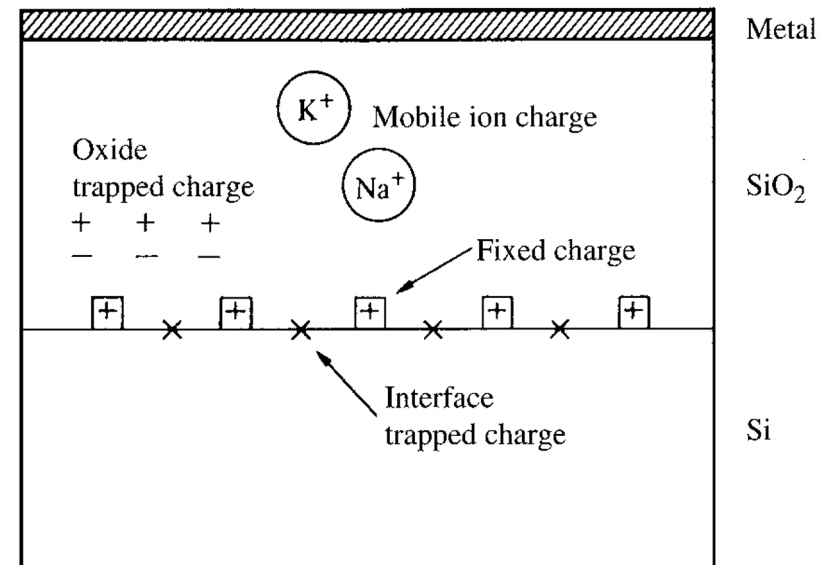
Threshold Voltage

– Oxide Charges –

With the types of oxide charges and their charge density

- (i) interface-trapped charges Q_{it} ,
- (ii) fixed oxide charges Q_f ,
- (iii) oxide-trapped charges Q_{ot} ,
- (iv) mobile ionic charges Q_m ,

the threshold voltage of the MOSFET becomes



Pierret, Fig. 18.4

$$V_T = \phi_{MS} + 2\phi_F - \frac{Q_B}{C_o} - \frac{Q_f}{C_o} - \frac{Q_{it}(\phi_s)}{C_o} - \frac{c_m Q_m}{C_o} - \frac{c_{ot} Q_{ot}}{C_o}$$

with $0 \leq c_m, c_{ot} \leq 1$; the charge density Q_{it} of the interface-trapped charges depends on the surface potential ϕ_s

Threshold Voltage

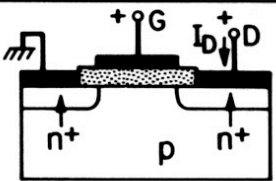
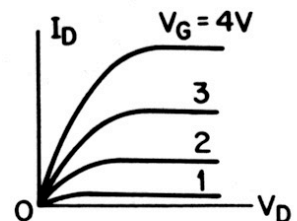
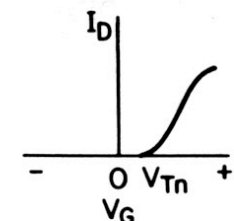
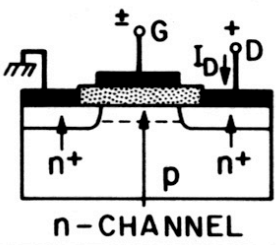
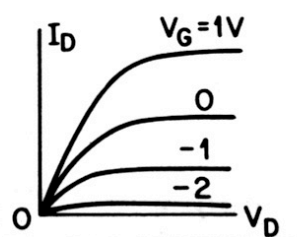
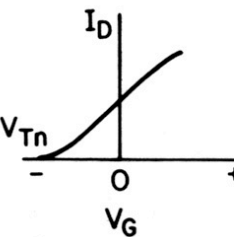
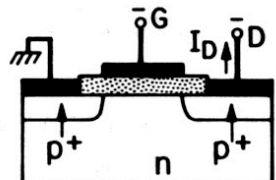
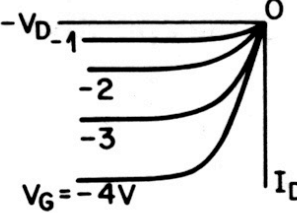
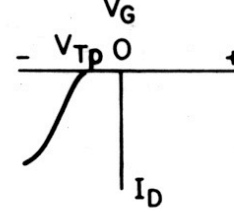
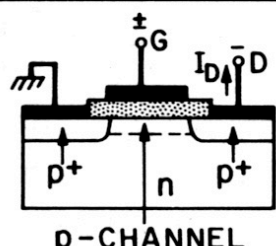
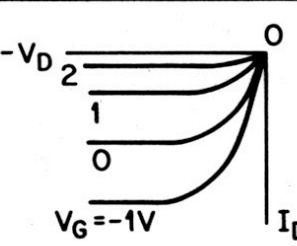
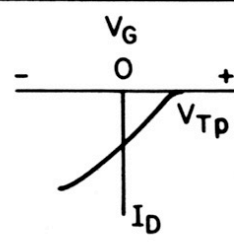
– Adjustment by Implantation –

- In modern device processing (e.g. n-channel and p-channel MOSFETs in CMOS processes), the **threshold voltage is adjusted to the desired value by ion implantation**, implanting a controlled number of either boron or phosphorous ions into the near-surface region of the semiconductor (in the channel region)
- Under inversion bias, the additional dopant adds to the charge near the Si-SiO₂ interface (with charge density Q_I) and, thus, results in a V_T shift of the structure:

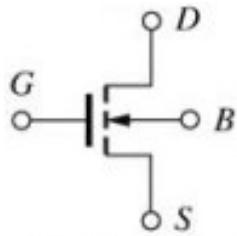
$$\Delta V_T = -Q_I / C_o$$

- Depending on the dopant type, both positive (for acceptor atoms, B) and negative (for donor atoms, P) V_T shifts are possible
- Example: $Q_I = 5 \cdot 10^{11} \text{ cm}^{-3}$ boron, $x_o = 10 \text{ nm}$ yields $\Delta V_T = +0.23 \text{ V}$

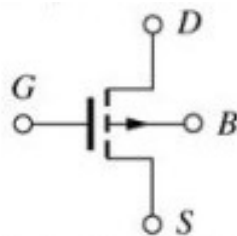
5.2.4 Types of MOSFET

TYPE	CROSS SECTION	OUTPUT CHARACTERISTICS	TRANSFER CHARACTERISTICS
n-CHANNEL ENHANCEMENT (NORMALLY OFF)			
n-CHANNEL DEPLETION (NORMALLY ON)			
p-CHANNEL ENHANCEMENT (NORMALLY OFF)			
p-CHANNEL DEPLETION (NORMALLY ON)			

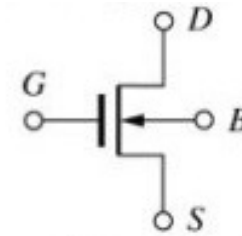
Sze, Fig. 6.19



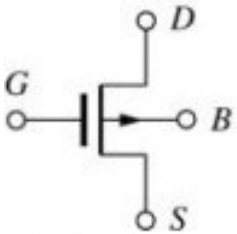
(a) NMOS enhancement-mode device



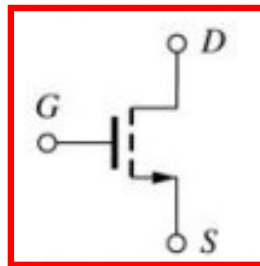
(b) PMOS enhancement-mode device



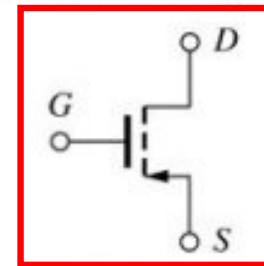
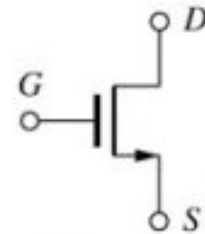
(c) NMOS depletion-mode device



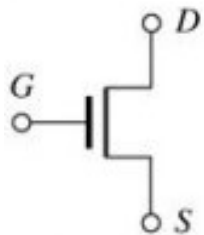
(d) PMOS depletion-mode device



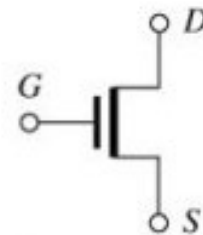
(e) Three-terminal NMOS transistors



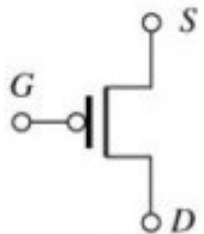
(f) Three-terminal PMOS transistors



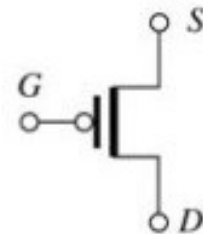
(g) Shorthand notation—NMOS enhancement-mode device



(h) Shorthand notation—NMOS depletion-mode device



(i) Shorthand notation—PMOS enhancement-mode device



(j) Shorthand notation—PMOS depletion-mode device

Jaeger, Blalock, Fig. 4.15

MOS Transistor Circuit Symbols